Anna Nariyeva

Atlanta, GA | (267) 267-7307 | anariyeva3@gatech.edu | linkedin.com/in/anna-nariyeva/

Portfolio

Career Objective

Mechanical Engineering student at Georgia Tech with a concentration in Manufacturing seeking a Summer 2027 engineering internship. Interested in hands-on roles involving mechanical design, manufacturing systems, automation, and aerospace production.

Education

Georgia Institute of Technology May. 2026 – Exp. May. 2028
B.S. in Mechanical Engineering, Concentration in Manufacturing — GPA: 3.50/4.00 *Atlanta, GA*

The Pennsylvania State University Jan. 2024 – May. 2026
B.S. in Mechanical Engineering, Minor in Mechatronics — GPA: 3.70/4.00 *Abington, PA*

Yonsei University — Korean Language Institute Aug. 2022 – Nov. 2023
Intensive Korean Course - Intermediate Level *Seoul, South Korea*

Engineering Experience

Part-time Undergraduate Biomechanical Research Assistant Jan. 2025 – May. 2026
The Pennsylvania State University *Abington, PA*

- Performed mechanical testing and data analysis of 100+ 3D-printed polypropylene orthotic braces using Tinius Olsen (tensile testing) equipment to quantify tensile, compressive, and torsional performance parameters on behalf of Korthotics, an Australian orthotics company.
- Analyzed experimental data to investigate the effects of print orientation, material type, extrusion temperature, and layer height on the strength and flexibility of 3D-printed components for ankle-foot orthoses (AFOs).
- Applied finite element analysis (FEA) in Fusion 360 and compared simulated and experimental results, calibrated material models, and improved the accuracy of digital simulations of additively-manufactured components.
- Presented findings at the ASB East Regional Conference at the University Delaware and collaborated with the biomechanical department to develop production methodologies that minimize the cost of manufacturing.

Full-time Biomechanical Research Assistant - Multi-campus REU May. 2025 – Aug. 2025
The Pennsylvania State University *University Park and Abington, PA*

- Quantified the flexural strength and stiffness of 20+ conventionally and commercially fabricated 3D-printed ankle-foot orthoses (AFOs) to establish performance benchmarks for clinically used devices.
- Eliminated testing inaccuracy by designing testing protocols and custom fixtures compatible with Tinius Olsen equipment to measure strength and deformation.
- Performed finite element analysis (FEA) to evaluate AFO designs and investigate design solutions that meet specified strength and performance requirements.
- Worked closely with Dr. Mukul Talaty through weekly research meetings to interpret experimental findings, refine testing methodologies, and develop independent problem-solving skills.

Part-time Engineering Laboratory Technician Sep. 2025 – May. 2026
The Pennsylvania State University *Abington, PA*

- Assembled, calibrated, and maintained 10+ 3D printers and mechanical testing equipment to support undergraduate engineering coursework and laboratory experiments.
- Trained and supervised 30+ undergraduate students in the safe operation of 3D printers and mechanical testing equipment, promoting laboratory safety and reliable experimental procedures.
- Troubleshot laboratory hardware and engineering software, including Prusa and Raise3D printers, SolidWorks, and Fusion 360, while assisting with experimental documentation.

Conferences

ASB East Regional Conference, University of Delaware April, 2026
Mechanical Performance of Polypropylene for FDM-Based 3D-Printed Ankle-Foot Orthoses

Multi-Campus Research in Engineering for Undergraduates (REU), Penn State University Park July, 2025
Comparing the Mechanical Properties for Conventionally Fabricated and 3D-Printed Ankle-Foot Orthoses

Skills

Engineering: CAD, Finite Element Analysis (FEA), Mechanical Testing, Additive Manufacturing, C++
Software: SolidWorks, Fusion 360, MATLAB, NI Multisim, Quartus Prime, Tinius Olsen Horizon
Manufacturing & Prototyping: FDM printing, SLS printing, FGF printing, PrusaSlicer, Cura
Languages: English (Native), Spanish (Native), Russian (Native), Korean (Intermediate)